TECHNICAL REPORT · TEAM COURSEWORK · M2 STATISTICS, SORBONNE UNIVERSITY · 2022-2023

Probabilistic Models: EM, Gibbs Sampling and Metropolis-Hastings

Latent-variable models and Bayesian inference implemented from scratch in R, and a review of Pseudo-Extended MCMC (Nemeth et al., NeurIPS 2019).

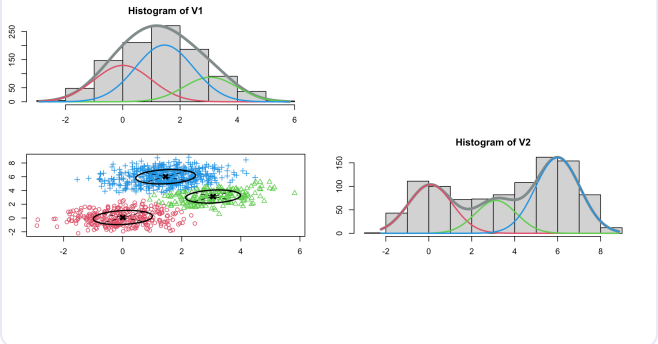
Code: github.com/6inzia/probabilistic-models-em-mcmc

1 EM algorithm for Gaussian mixtures

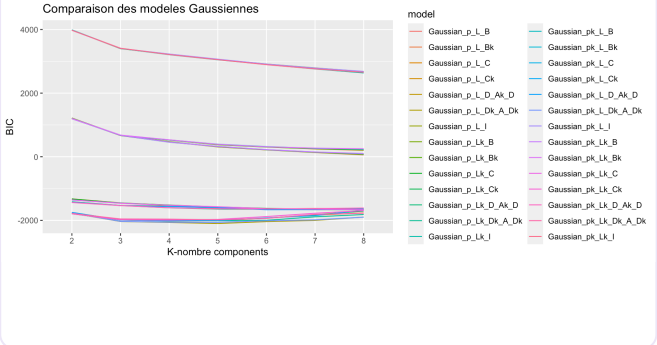
For a mixture $f(x) = \sum_{k=1}^{K} p_k \, \varphi(x; \mu_k, \sigma_k^2)$ with latent labels z_i, the EM algorithm maximizes the likelihood by alternating two closed-form steps until the log-likelihood stops increasing:

- **E-step**: posterior probabilities $t_{ik} = P(z_i = k \mid x_i, \theta) = \dfrac{p_k \varphi(x_i; \mu_k, \sigma_k^2)}{\sum_l p_l \varphi(x_i; \mu_l, \sigma_l^2)}$
- **M-step**: $p_k = \frac{1}{n} \sum_i t_{ik}$, $\mu_k = \dfrac{\sum_i t_{ik} x_i}{\sum_i t_{ik}}$, $\sigma_k^2 = \dfrac{\sum_i t_{ik} (x_i - \mu_k)^2}{\sum_i t_{ik}}$

Observations are then classified by the maximum a posteriori rule, and the number of components K is chosen by **BIC** $= -2 \log L + \nu \log n$, which penalizes the number of parameters ν. In higher dimension, model-based clustering was studied with Rmixmod: 14 covariance structures (spherical, diagonal, general) compared with BIC and ICL.



2D mixture of 3 Gaussians: the 3 components and their covariance ellipses are recovered.



BIC of every covariance model as a function of K (seeds data set): the "general" family is selected.

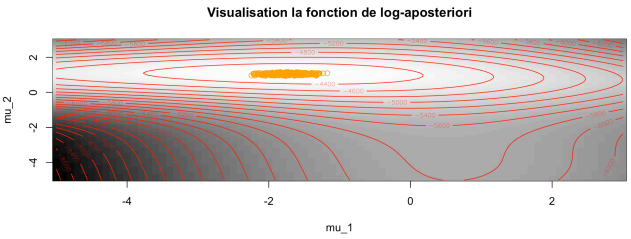
Findings. With well-separated components, EM recovers the proportions, means and variances accurately and the log-likelihood stabilizes within a few iterations; with more components it can plateau before increasing again. When components overlap and samples are small, BIC tends to merge components (7 simulated Gaussians → 4 selected), and in high dimension with few observations it can even split a single Gaussian into several clusters.

2 Gibbs sampler for a Bayesian Gaussian mixture

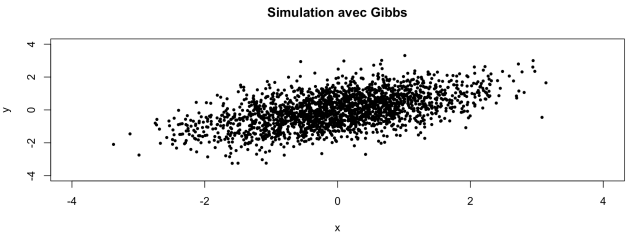
We place conjugate priors on the parameters: $\pi \sim \mathcal{D}(\gamma_1, \gamma_2)$, $\sigma_k^2 \sim \mathcal{IG}(\lambda_k/2, \beta_k/2)$ and $\mu_k \mid \sigma_k^2 \sim \mathcal{N}(\alpha_k, \sigma_k^2/\lambda_k)$. The posterior is intractable, but both full conditionals are easy to sample (**data augmentation**):

- $z_i \mid x, \theta$: categorical with probabilities $\propto \pi_k \, \varphi(x_i; \mu_k, \sigma_k^2)$;
- $\theta \mid z, x$: Dirichlet, inverse-gamma and Gaussian updates using the counts, sums and sums of squares of each group.

After a burn-in of 500 iterations, the posterior means are very close to the true values ($\hat{\mu}_1 \approx -1.6$ vs -1.58, $\hat{\mu}_2 \approx 1.0$ vs 0.99). The same principle samples a correlated Gaussian from its conditionals $x \mid y \sim \mathcal{N}(\rho y, 1 - \rho^2)$ and $y \mid x \sim \mathcal{N}(\rho x, 1 - \rho^2)$.



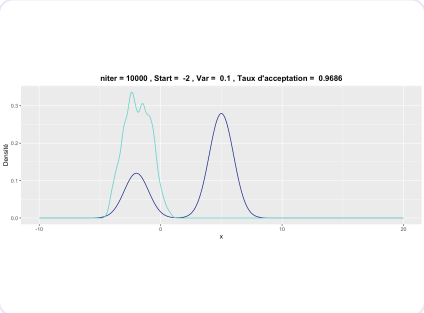
Log-posterior of (μ_1, μ_2) with the Gibbs samples (orange) concentrated on its mode.



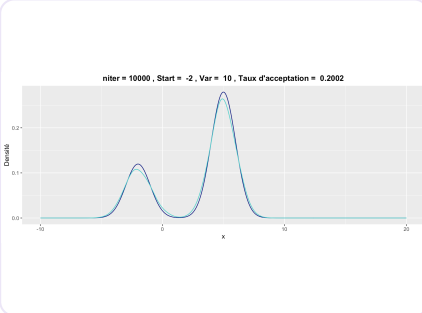
Gibbs sampling of a bivariate Gaussian with $\rho = 0.5$.

3 Metropolis-Hastings and tempering

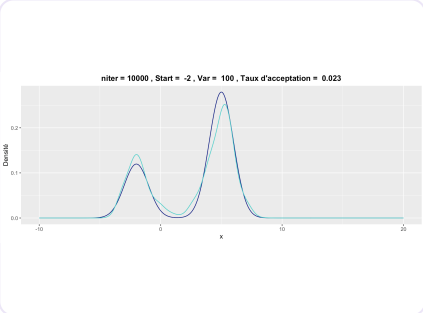
Random-walk Metropolis-Hastings proposes $\tilde{x} \sim \mathcal{N}(x_{t-1}, \zeta^2)$ and accepts it with probability $\min\{1, \, \pi(\tilde{x}) / \pi(x_{t-1})\}$, so only an unnormalized density is needed. On the bimodal target $0.3\,\mathcal{N}(-2, 1) + 0.7\,\mathcal{N}(5, 1)$, the scale ζ drives everything:



$\zeta = 0.1$: 97% accepted, the chain stays in one mode.

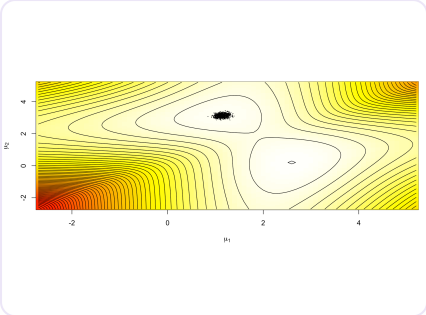


$\zeta = 10$: 20% accepted, both modes are explored.

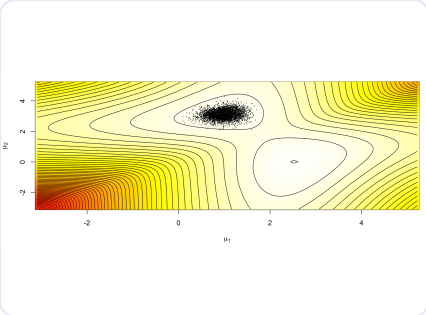


$\zeta = 100$: 2% accepted, too many rejections.

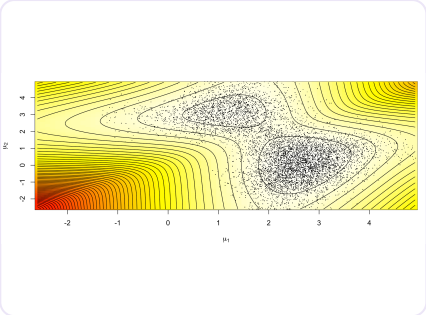
Tempering. For the posterior of the means of a mixture, a chain started next to a spurious mode stays trapped. Targeting π^α with $\alpha < 1$ flattens the surface (lower peaks, higher valleys): with $\alpha = 0.01$, the chain reaches the main mode within a few iterations, while with $\alpha = 1$, 10^4 iterations are not enough.



$\alpha = 1$: stuck in the spurious mode.



$\alpha = 0.1$: starts to spread.



$\alpha = 0.01$: moves between modes.

Comparison. On the same mixture, the Gibbs sampler gave more precise estimates of the means than Metropolis-Hastings, which needed about 20 iterations to forget a poor starting point; EM reached the maximum likelihood in about 5 iterations, but only gives a point estimate, without posterior uncertainty.

4 Research review: Pseudo-Extended MCMC (NeurIPS 2019)

Problem. MCMC samplers can stay trapped in a local mode of a multimodal posterior for a very long time (mixture models, deep neural networks), exactly as observed above.

Idea. The state space is extended with N pseudo-samples. With $\pi(x) = \gamma(x)/Z$ and an auxiliary density q, the extended target is

$$\pi^N(x_{1:N}) = \frac{1}{Z}\left\{\frac{1}{N}\sum_{i=1}^{N}\frac{\gamma(x_i)}{q(x_i)}\right\}\prod_{j=1}^{N} q(x_j).$$

On this space the modes are connected, so Hamiltonian Monte Carlo moves easily between them. Unlike pseudo-marginal MCMC, q only needs to be **evaluated**, not sampled, and neither the number of modes nor a temperature ladder has to be known.

Main result. Self-normalized importance weights give unbiased estimates under the original target:

$$\mathbb{E}_{\pi^N}\left[\frac{\sum_i f(x_i)\,\gamma(x_i)/q(x_i)}{\sum_i \gamma(x_i)/q(x_i)}\right] = \mathbb{E}_{\pi}[f(x)].$$

Limitations. Sensitivity to the hyperparameters (temperature, step size, number of iterations) and a computational cost that grows with N; for simple unimodal targets, standard HMC remains preferable.

> Gibbs sampler adapted from the course notes of the "Latent structure models" class (Sorbonne University).
> Tempering experiment from J.-M. Marin & C. P. Robert, *Bayesian Essentials with R* (Springer, 2014).